

Given: Due:	Further Maths (Core Pure) HW 18	Name:
Chapter 3: Series		
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.		
91% (A*)	76% (A)	66% (B)
56% (C)	50% (D)	40% (E)
<i>Useful resources:</i> Your notes from recent lessons & your text book <u>Use your formula book! Use the Σ function on your calculator to check results where possible (numeric values). This topic is algebra intensive so work carefully, one step at a time, line by line, and double check your solution.</u>		
Questions - do these on lined paper		
Q1. (a) Use the standard results for $\sum_{r=1}^n r^3$ and $\sum_{r=1}^n r$ to show that $\sum_{r=1}^n (r^3 + 6r - 3) = \frac{1}{4}n^2(n^2 + 2n + 13)$ for all positive integers n. (5) (b) Hence find the exact value of $\sum_{r=16}^{30} (r^3 + 6r - 3)$ (2)		
Q2. (a) Show, using the formulae for $\sum r$ and $\sum r^2$, that $\sum_{r=1}^n (6r^2 + 4r - 1) = n(n+2)(2n+1)$ (5) (b) Hence, or otherwise, find the value of $\sum_{r=1}^{20} (6r^2 + 4r - 1)$. (2)		
Q3. (a) Using the formula for $\sum_{r=1}^n r^2$ write down, in terms of n only, an expression for $\sum_{r=1}^{3n} r^2$ (1) (b) Show that, for all integers n, where $n > 0$ $\sum_{r=2n+1}^{3n} r^2 = \frac{n}{6}(an^2 + bn + c)$ where the values of the constants a, b and c are to be found. (4)		



Q4.

- (a) Using the formulae for $\sum_{r=1}^n r$ and $\sum_{r=1}^n r^2$, show that

$$\sum_{r=1}^n (r+1)(r+4) = \frac{n}{3}(n+4)(n+5)$$

for all positive integers n .

(5)

- (b) Hence show that

$$\sum_{r=n+1}^{2n} (r+1)(r+4) = \frac{n}{3}(n+1)(an+b)$$

where a and b are integers to be found.

(3)

Q5.

- (a) Use the results for $\sum_{r=1}^n r$ and $\sum_{r=1}^n r^2$ to show that

$$\sum_{r=1}^n (2r-1)^2 = \frac{1}{3}n(2n+1)(2n-1)$$

for all positive integers n .

(6)

- (b) Hence show that

$$\sum_{r=n+1}^{3n} (2r-1)^2 = \frac{2}{3}n(an^2+b)$$

where a and b are integers to be found.

(4)

END OF HOMEWORK**Total = 37 marks**

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.